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Project Title:	Molecular Genetics of Root Thigmoresponsiveness in Arabidopsis thaliana		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
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Zip Code:	53706-1580	**Congressional District:**	2
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No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
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Contact Monitor:		**Contact Phone:**	
Contact Email:			
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When Arabidopsis thaliana seedlings grow on tilted agar surfaces, their roots develop a wavy pattern of growth in response to gravity, touch and other surface-derived stimuli (Okada and Shimura, 1990, Science 250: 274-276). To better understand that process, we have identified mutants affected in the ability of their roots to wave on tilted agar surfaces. Some of them show specific defects in root gravitropism, while others are normally gravitropic. The latter are likely to be affected in sensing or transducing touch and/or other surface-derived stimuli responsible for root waving, or to have structural abnormalities interfering with their ability to wave. The specific objectives of this research proposal are to characterize two genes (WVD2 and WVD6) essential for root waving and skewing on tilted agar surfaces. wvd2 and wvd6 mutant seedlings are characterized by dampened root waves on tilted agar surfaces. Furthermore, wvd2 roots skew in the opposite direction compared to wild-type when growing on vertical agar surfaces, while wvd6 roots do not skew. We propose to clone and sequence the WVD2 and WVD6 genes, to define their expression patterns under induced

Task Description:	and uninduced conditions, as well as in various root-wave and root-gravitropism mutants, to define the cellular and subcellular location of the corresponding proteins, and to better characterize the genetic, physiological, and morphological properties of the mutants in order to better understand functions of the corresponding genes. The proposed research, along with the genetic and molecular analysis of other loci involved in root waving, will allow us to better understand how mechanical stimuli affect the pattern of root growth. These studies are of great relevance to the NASA space biology program for the efficiency by which roots use environmental cues to direct their growth will condition plant growth and production in microgravity environments. The objective of this research is to characterize two genes, WVD2 and WVD6, that were previously implicated in the control of root growth behavior in response to mechanical stimulation, in the hope that the corresponding information will provide new insights into the mechanisms that govern thigmoresponsiveness in plants. The WVD2 and WVD6 genes appear to function not only as key regulators of root growth behavior in response to mechanical stimuli, but also as major determinants in the control of anisotropic cell expansion in all organs of the plant. Therefore, a better understanding of these genes is likely to provide important new insights into how mechanical information in the environment controls anisotropic cell expansion in plant organs.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Root-growth behavior conditions the ability of plants to populate and grow in very heterogenous and diverse environments. Hence, a better understanding of the molecular mechanisms that govern this process is likely to provide the tools needed to engineer plants that are better able to colonize extreme environments on Earth as well as on other planets. At the same time, controlled anisotropic cell expansion is an essential component of plant morphogenesis and behavior in response to directional environmental stimuli. Our studies have allowed the identification of an important new dominant regulator of this process. Because over-expression of WVD2 results in shorter plants with thicker organs, it may be possible to use this engineering strategy to generate plants that are more resistant to lodging by wind and rain, as well as plants that are better suited for utilization in spacecrafts, where limited amounts of space are available for plant growth.
Task Progress:	When Arabidopsis thaliana seedlings are forced to grow on tilted hard agar surfaces, their roots tend to grow down the plate following a wavy pattern that results from a combination of gravity and touch stimulation, along with some contribution from endogenous processes (circumnutation). These roots also tend to skew their growth to the right of gravity. To better understand the molecular mechanisms that underlie root growth behavior in response to mechanical stimulation, we previously reported on the identification of wvd2-1 and wvd6-1, two mutations that affect root waving on tilted hard agar surfaces, and either invert (wvd2-1) or eliminate (wvd6-1) root skewing on such surfaces. We showed that wvd2-1 results from over-expression of a small wild type protein, whereas wvd6-1 is a knockout mutation of RHD3, a gene previously implicated in root hair development and anisotropic cell expansion. Last year, we continued our analysis of wvd2-1 and wvd6-1, uncovering important mechanisms that might modulate their respective functions in the regulation of root growth behavior and anisotropic cell expansion. We showed that the WVD2 protein carries a conserved domain found in proteins that interact with microtubules in animal cells. Using this information, we further demonstrated that WVD2 co-fractionates with the cytoskeleton in biochemical fractionation assays. Importantly, WVD2 was shown to display a strong in vitro microtubule bundling activity, suggesting a biochemical mechanism underlying its role in the control of anisotropic cell expansion and root growth behavior. Consistent with this, cortical microtubules in rapidly expanding cells of wvd2-1 seedlings displayed a strong alteration in cortical microtubule alignment compared to wild type. In vivo analyses of microtubule dynamics and WVD2 protein localization within plant cells are underway to test this important model. Furthermore, proteins that interact with WVD2 in a yeast two-hybrid system, and genetic modifiers of wvd2-1 have been identified and are being characterized. Finally, several genes encoding WVD2-like proteins have been identified, and their functions are being characterized. Our analysis of wvd6-1 also allowed us to demonstrate that RHD3 regulates spiral root growth in Arabidopsis in a specific fashion, distinct from its role in root hair development. Its contribution to spiral root growth may be associated with its ability to control the deposition of wall components, or to control the organization of actin microfilaments in rapidly expanding cells. To summarize, our research on WVD2 and WVD6 has allowed us to discover a possible role for microtubule bundling and wall material deposition or microfilament organization in controlling spiral root growth, root waving and anisotropic cell expansion in plants.
Bibliography Type:	Description: (Last Updated: 07/03/2024)
Articles in Peer-reviewed Journals	Blancaflor EB, Masson PH. "Plant gravitropism. Unraveling the ups and downs of a complex process." Plant Physiol. 2003 Dec;133(4):1677-90. Review. https://doi.org/10.1104/pp.103.032169 ; PubMed PMID: 14681531; PubMed Central PMCID: PMC1540347 , Dec-2003
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Dissertations and Theses	Yuen CYL. "A Genetic Analysis of Spiral Root Growth in Arabidopsis thaliana: Characterization of wvd2-1 and wvd6." PhD Thesis, Genetics Program, University of Wisconsin – Madison, June 2003. , Jun-2003